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imud — Mathematical Reference (as implemented)

Document purpose. This is an audit reference for the *actual numerical methods executed by the imud source code*, at the release this file ships with. It is deliberately **not** a derivation of the methods imud “should” use, nor a restatement of the papers it draws on: every equation below was transcribed from the code and is annotated with the function, variable, and struct-field names so a reviewer can place each symbol against its implementation. Where the code approximates, deviates from, or specializes its cited source, an **As-implemented** note says so explicitly.

Typeset copy. `docs/math.pdf` in the source repository renders these equations properly for reading away from a terminal. It is committed but not packaged — it would only duplicate this file — and it is not linked from here, because a relative link to it would be dead in the installed doc tree. It is kept honest by `make check-math-pdf-stamp`, which compares it against this file by hash; after editing here, run `make math-pdf` and commit both. That needs `pandoc` and either `tectonic` (a single binary) or a XeLaTeX.

Intended reader. Someone auditing the mathematics itself — an estimation or geophysics specialist — who wants to check the equations and their data flow without reading C. Familiarity with quaternion attitude estimation, Kalman filtering, and spherical-harmonic geomagnetism is assumed.

Scope. All numeric routines in the daemon and its calibration tool: the MEKF attitude/heading filter, accelerometer/magnetometer measurement models, the heave and sea-state estimators, compass-health and engine-vibration metrics, timestamp anchoring, the offline magnetometer / Allan-variance / gyro-temperature calibration fits, and the World Magnetic Model evaluation. The WMM section states the *implementation shape* and cites the defining report rather than re-deriving spherical-harmonic theory.

Source-of-truth files. `src/fusion.c`, `include/fusion.h` (MEKF, heave, sea state); `src/imu.c` (live calibration application, Euler rates, rate of turn, timestamp anchor, engine detector, startup bias); `src/cal_math.c`, `include/cal_math.h` (offline fits, Allan variance); `src/wmm.c`, `include/wmm.h` (geomagnetic model); `src/cal.c` (calibration file I/O).

Citation convention. Each section ends with a **Source** line. Because imud implements standard methods rather than novel mathematics, most sources are the canonical reference for the technique. A citation marked (*code comment*) is the reference named in the source itself; one marked (*canonical*) is the standard literature reference for a technique the code does not cite in-line. See §15 References. Citations the maintainer may wish to verify against the exact edition are flagged in that section.

1. Conventions and reference frames

Frames.

- **Body frame b**: the sensor board axes after the optional mount rotation (§3). NED-compatible: X forward, Y starboard, Z down.
- **Navigation frame NED**: local North-East-Down.

Attitude quaternion. `mekf_t.q = q = [qw, qx, qy, qz]` is a unit Hamilton quaternion, **scalar-first**, representing the **body→NED** rotation:

$$v_{NED} = R(q) v_b, \quad R(q) \in SO(3).$$

The rotation matrix `q_to_R()` (`fusion.c:280`) is

$$R(q) = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z - q_w q_x) \\ 2(q_x q_z - q_w q_y) & 2(q_y q_z + q_w q_x) & 1 - 2(q_x^2 + q_y^2) \end{bmatrix}.$$

The Hamilton product `q_mul()` (`c = a ⊗ b`, `fusion.c:243`) uses the standard scalar-first convention (Solà eq. 16).

Units. Gyroscope $\text{rad} \cdot \text{s}^{-1}$; accelerometer $\text{m} \cdot \text{s}^{-2}$; magnetometer μT on the wire, converted to **Gauss** ($\times 0.01$) inside the filter; angles rad ; time s . Standard gravity $\mathbf{G_MS2} = g = 9.80665 \text{ m s}^{-2}$ (`fusion.c:43`).

Gravity reference. $g_{ref} = [0, 0, 1]$ (unit, NED, Z-down). The accelerometer's *specific force* at rest reads $-g$ on Z; the filter works with the **gravity direction** $z_a = -\hat{a}_b$ (§4.7).

As-implemented. MEKF state — quaternion, bias, wave acceleration, and the covariance P — is single precision (`float`); calibration fits and WMM are double precision. The quaternion is renormalized (`q_normalize`, `fusion.c:266`) after every multiplicative update, guarding unit-norm drift from float round-off, and P uses the Joseph form for the same reason (§4.5): the simple $(I - KH)P$ form slowly loses symmetry and positive-definiteness at 833 Hz over multi-day runs.

Where that split stops holding, and why the accumulators are double. The rule is not “filter state is float” but *nothing that accumulates over an unbounded number of steps is float*. Every leaky integrator and exponentially weighted statistic here advances by $\alpha \delta$ with $\alpha = \Delta t / \tau$, and that gain shrinks with the sample rate **and** with the time constant. Once α falls below half a float32 ULP of the state it is being added to, the update rounds to an exact no-op — not a gradual loss, a full stop — and $\varepsilon_{32} = 1.19 \times 10^{-7}$ is reachable from `imud.conf`: at 32 kHz a 1200 s sea-state window gives $\alpha = 2.6 \times 10^{-8}$.

Measured against a double-precision reference of the same arithmetic, both fed identical float-rounded inputs so only accumulator width differs:

```
rm -f test_fusion && make test_fusion CFLAGS="-D_GNU_SOURCE -O2 -Wall \
-Wextra -std=c11 -pthread -Iinclude -DBENCH_SWEEP_PRECISION" && ./test_fusion
```

$\Delta t / \tau$	configuration	Hs error	T_z error
1.0×10^{-5}	833 Hz / 120 s (default)	0.000 %	0.000 %
2.1×10^{-7}	8 kHz / 600 s	0.128 %	0.491 %
5.2×10^{-8}	32 kHz / 600 s	1.823 %	0.982 %
2.6×10^{-8}	32 kHz / 1200 s	17.995 %	6.865 %

Heave, on its own τ axis, failed harder. Its leak is what bounds the drift of a double integration of accelerometer noise; when the leak becomes a no-op the estimator is a *plain* double integrator:

$\Delta t/\tau$	configuration	recovered amplitude error
2.6×10^{-6}	32 kHz / 12 s (default)	0.225 %
5.2×10^{-7}	32 kHz / 60 s	1.225 %
1.0×10^{-7}	32 kHz / 300 s	7.263 %
3.5×10^{-8}	32 kHz / 900 s	6 543 975 % — 253 m of "heave"

Both estimators' accumulators are therefore **double** (**seastate_t**'s six mean/variance pairs, **heave_t**'s **vel/disp/hp_y**), along with their gains and heave's gravity subtraction — a cancellation of 9.8 against 9.8 down to a signal three orders smaller. That moves the cliff to $\Delta t/\tau \sim 10^{-16}$, which no rate and window reach. Every cell above reads 0.000 % after. Inputs, outputs and the wire stay float: sensor resolution is coarser than float32 either way, so widening those would flatter the measurement without changing the hardware.

Independent confirmation from the default-on sweeps: **test_seastate_across_rates** was flat at 1.015 % Hs from 100 Hz to 16 kHz and ticked up to 1.046 % at 32 kHz. §4.7.2 recorded that endpoint as float32 accumulation. It is now 1.016 % — the plateau simply continues, and the uptick is gone.

One elapsed-time gain is worth naming separately. **mekf_ema_alpha** ($\alpha = 1 - e^{-\Delta t/\tau}$, exact for any feed rate) must be computed with **expm1f**: spelled **1.0f - expf(-x)** it subtracts two numbers differing by less than an ULP of 1.0, quantising the answer to multiples of 5.96×10^{-8} — 2.7 % error at 32 kHz against the 30 s health constant, with only 17 representable values.

Source: Solà 2017 (*code comment*); Shuster 1993 for quaternion conventions (*canonical*); Higham 2002 for the $1 - e^{-x}$ cancellation (*canonical*). Measurements from **-DBENCH_SWEEP_PRECISION**.

2. Signal flow

One IMU sample traverses, in order:

1. **Driver read** $\rightarrow$ **raw_imu_sample_t**{**accel**[3], **gyro**[3], **temp_c**, **chip_ts**} in board axes.
2. **Mount rotation** **apply_mount_rot_if_set()** (§3.1) $\rightarrow$ body axes.
3. **Calibration** **apply_imu_cal()** (§3.2): gyro temperature compensation, accel offset/scale. (Magnetometer: **apply_mag_cal()**, hard/soft iron.)
4. **MEKF** (§4): **mekf_predict** on every IMU sample; **mekf_update_accel** every sample; **mekf_update_mag** per magnetometer sample.
5. **Derived outputs**: Euler angles and heading (§4.10); Euler rates and rate of turn (§5); heave (§6); sea-state statistics (§7); compass-health (§8) and engine (§9) metrics.

Gyro **bias** is *not* removed in step 3; the MEKF estimates and subtracts it (§4.4). Timestamps are reconstructed from the chip counter (§11).

3. Calibration application (live path)

3.1 Mount rotation

apply_mount_rot_if_set() (**imu_math.c**:435) applies a fixed 3×3 board $\rightarrow$ body matrix $R_{mount} = \text{cfg} \rightarrow \text{mount_rot}$ in place, when **cfg $\rightarrow$ mount_set**:

$$v \leftarrow R_{mount} v, \quad v \in \{a_b, \omega_b, m_b\}.$$

Applied identically to accel, gyro, and magnetometer vectors. Accumulated in double, stored back as float.

As-implemented. R_{mount} is supplied by configuration (a fixed installation rotation, e.g. yaw 180° for a stern-facing board); `imu` does not estimate it. It may be given as Euler angles (`rotation_euler_deg`), a named `preset`, or directly as a 3×3 `rotation_matrix`. The first two are orthonormal by construction; a directly-supplied matrix is validated at config load against both $R^\top R \approx I$ and $\det R > 0$ (rejecting reflections, which an orthogonality-only test would accept) and the daemon refuses to start if it fails.

3.2 Inertial calibration

`apply_imu_cal()` (`imu_math.c:26`), per axis i :

- **Gyro temperature compensation** (when `cal->has_gyro_temp`):

$$\omega_i \leftarrow \omega_i - c_i (T - T_{ref}),$$

with $c_i = \text{gyro_temp_coeff}[i]$, $T = \text{temp_c}$, $T_{ref} = \text{gyro_temp_ref_c}$ (fit in §12.6).

- **Accel offset/scale** (when `cal->has_accel`):

$$a_i \leftarrow (a_i - o_i) s_i,$$

with $o_i = \text{accel_offset}[i]$, $s_i = \text{accel_scale}[i]$.

3.3 Magnetometer hard/soft-iron

`apply_mag_cal()` (`imu_math.c:44`), when `cal->has_mag`:

$$m_{cal} = S_{soft} (m_{raw} - b_{hard}),$$

with $b_{hard} = \text{mag_hard_iron}[3]$, $S_{soft} = \text{mag_soft_iron}[3][3]$. The correction parameters are produced offline (§12.2–12.3).

As-implemented. The accel model is diagonal scale + offset (no off-diagonal cross-axis / misalignment terms). The magnetometer model is a full 3×3 soft-iron matrix times a hard-iron offset, but that matrix is fitted from a **2-D** swing (§12.3), so its out-of-plane (dip) accuracy is limited — which is exactly why the filter defaults to heading-only magnetometer fusion (§4.8).

Source: standard sensor error model; Titterton & Weston 2004 §12 (*canonical*).

4. Multiplicative Extended Kalman Filter

The attitude/heading core is an **error-state (multiplicative) EKF**. The nominal state carries the full quaternion and gyro bias; the Kalman filter operates on a minimal 6-D **error state**, and corrections are injected multiplicatively so the quaternion never leaves the unit sphere.

4.1 State and covariance

Nominal state (`mekf_t`, `fusion.h:46`):

- q — unit quaternion, body→NED (`q[4]`).
- b — gyro bias, $\text{rad} \cdot \text{s}^{-1}$ (`bias[3]`).
- a_w — wave acceleration, body frame, normalized gravity units (`wave_acc[3]`); see §4.1.1.

Error state (never stored explicitly; the axis of P):

$$\delta x = [\delta\theta(3) \mid \delta b(3) \mid \delta a_w(3)] \in \mathbb{R}^9,$$

$\delta\theta$ = small-angle rotation error (rad), δb = bias error, δa_w = wave-acceleration error. Covariance $P \in \mathbb{R}^{9 \times 9}$ ($P[\text{MEKF_N}][\text{MEKF_N}]$, $\text{MEKF_N} = 9$), symmetric PD; top-left 3×3 is attitude-error covariance (rad^2), the middle 3×3 is bias-error covariance ($(\text{rad} \cdot \text{s}^{-1})^2$), the trailing 3×3 is wave-acceleration-error covariance (g^2).

The wave block is **appended**, not inserted, so the attitude and bias blocks keep their indices: `mekf_get_state()` reads $P[0:3][0:3]$ and $P[3][3]$, $P[4][4]$, $P[5][5]$ exactly as before and the wire format is unchanged.

4.1.1 The wave-acceleration state The gravity measurement is contaminated in a seaway by wave-orbital acceleration, which is **correlated over ~0.5–1.5 s, not white**. A white R_a therefore tells the filter that 833 samples per second are 833 independent measurements of gravity when they are closer to one per wave. The consequence is not subtle: the filter's covariance collapses, its gain vanishes, and it diverges into its own over-confidence (measured: attitude RMS $9.5^\circ/11.4^\circ$, NIS ≈ 56 — §4.7).

a_w is modelled as a **first-order Gauss–Markov (Ornstein–Uhlenbeck) process** with steady-state standard deviation σ and correlation time τ :

$$\dot{a}_w = -\frac{1}{\tau}a_w + w, \quad \mathbb{E}[a_w a_w^\top] = \sigma^2 I_3.$$

σ is configured in $\text{m} \cdot \text{s}^{-2}$ (`mekf_wave_accel`, the units a spec sheet and `imud-cal fit-ra` both speak) and stored as the variance $\sigma_g^2 = (\sigma/g)^2$ in `wave_sig2`; τ is `mekf_wave_accel_tau_s`. Either at 0 disables the state, and the disabled filter is bit-for-bit the pre-1.7 6-state filter — P 's wave rows and columns stay identically zero, Φ 's block is I , and the measurement Jacobian's block is never executed.

Frame. a_w is carried in the **body** frame. Wave-orbital acceleration is arguably a property of the seaway and so NED-stationary, which would add a transport term $\Phi_w = e^{-dt/\tau}(I - [\omega dt]_\times)$. That was implemented and measured (`-DWAVE_TRANSPORT` in `fusion.c`): it changes the benchmark in the third decimal and is slightly *worse* on yaw-only attitude RMS ($2.308^\circ \rightarrow 2.325^\circ$), because at $\pm 15^\circ$ of roll the frame rotation is small compared with τ . The body-frame form is kept for the simpler Jacobian.

Observability. $\delta\theta$ and δa_w both act in the tangent plane of the measured direction, so they are separated *only* by their dynamics — attitude error is a random walk driven by gyro noise, wave acceleration decays with τ . Too large a σ or τ lets the wave state absorb genuine tilt error; §4.7 records the sweep that bounds this.

The nominal/error relationship is the **right** (local) perturbation $q_{true} = \hat{q} \otimes \delta q(\delta\theta)$, established by the sign of the measurement Jacobian in §4.5.

Source: Solà 2017 §5.4 (*code comment*); Markley & Crassidis 2014 §6.1; Trawny & Roumeliotis 2005 (*canonical*).

4.2 Initialization — `mekf_init()` (`fusion.c:1019`)

Nominal: $q = [1, 0, 0, 0]$, $b = \text{gyro_bias_init}$ (from the startup still window, §10; may be 0).

Initial covariance (diagonal):

$$P_0[0:3] = (0.175)^2 \text{ rad}^2 (\approx 10^\circ), \quad P_0[3:6] = (0.001)^2 (\text{rad s}^{-1})^2, \quad P_0[6:9] = \sigma_g^2.$$

The wave block starts at its **steady state**, not at a wide acquisition value: the Gauss–Markov process is stationary, so there is no transient to model. (It is seeded to 0 when the state is disabled, which is what keeps the block inert.)

Discrete process-noise variances, step $dt = 1/\text{ODR}$:

$$Q_g = N_g^2 dt, \quad Q_b = N_b^2 dt,$$

with $N_g = \text{mekf_gyro_noise}$, $N_b = \text{mekf_gyro_bias}$ (datasheet noise densities). Stored as `f->Qg`, `f->Qb`.

Measurement-noise variances:

$$R_a = \left(\frac{N_a}{g} \right)^2 \text{ ODR}, \quad R_m = N_m^2 f_{s,\text{mag}},$$

f->Ra (mekf_derive_tuning, fusion.c:897) in normalized (gravity-direction) units, f->Rm (:898) in Gauss², with $N_a = \text{mekf_accel_noise}$, $N_m = \text{mekf_mag_noise}$, $f_{s,\text{mag}} = \text{mag_odr_hz}$.

Derived thresholds: accel skip band $[1 - s_k, 1 + s_k]$ with $s_k = \text{accel_skip_thresh}$; $\text{mag_reject_sq} = (\text{mag_reject_gauss})^2$; convergence threshold $\text{conv_thresh} = 3\theta_c^2$ with $\theta_c = \max(0.5^\circ, 0.30 \arcsin \sigma_g)$ — 1.40° at the default σ . It scales with σ_g rather than being flat because with the wave state P reports a believable steady-state attitude variance, and that has a floor near $0.9^\circ/\text{axis}$: a fixed 0.5° threshold would never be reached (§4.1.1). m_ref EMA gain $mref_alpha = 1/(\tau_{mref} f_{s,\text{mag}})$ with $\tau_{mref} = 300 \text{ s}$ (:941).

As-implemented. Q_g, Q_b are *per nominal step*; during prediction they are rescaled by the actual dt (§4.4). The bias process is modeled as a random walk whose density N_b is a **tuning constant deliberately held above** the measured in-run bias instability (see docs/capture.md); it is not driven by the Allan-variance characterization of §12.5.

Source: Solà 2017 §4.1 for discrete process noise (*code comment*).

4.3 Alignment — mekf_align() (fusion.c:1077)

Deterministic initial attitude from one static accel+mag pair (a tilt-then-heading decomposition, TRIAD-family).

The caller supplies a window mean, not one sample (imu.c, the align loop): the accelerometer and magnetometer are averaged over `align_window_sec` before this is called. That window matters far more than its obscurity suggests, because whatever tilt error survives it is baked permanently into m_{ref} 's dip (§4.8.1) and, in 3-D mode, becomes a constant roll/pitch bias. One second — the hardcoded value before 1.7 — is about a fifth of a typical roll period, so it averages an arbitrary fraction of the cycle. Measured over the 12-seed wave benchmark, attitude RMS:

window	yaw-only (default)	3-D	3-D NEES(strict)
1 s	47.7°	6.72°	339
2 s	2.28°	5.78°	250
3 s	2.11°	2.38°	38.9
5 s (default)	2.19°	1.18°	5.74
15 s	2.19°	0.90°	1.71
30 s	2.24°	0.93°	1.94

The marine default is flat from ~2 s; 3-D keeps improving to ~15 s. The default is 5 s, and the cost of a longer window is purely startup latency before the filter produces usable attitude — worth paying at a mooring, not underway.

Tilt from accelerometer. With $\hat{g}_b = -a_b/\|a_b\|$ the gravity direction in body (guard: reject if $\|a_b\| < 0.5g$):

$$\phi = \text{atan2}(\hat{g}_{b,y}, \hat{g}_{b,z}), \quad \theta = \text{atan2}(-\hat{g}_{b,x}, \sqrt{\hat{g}_{b,y}^2 + \hat{g}_{b,z}^2}).$$

Tilt quaternion $q_{\text{tilt}} = q_\theta \otimes q_\phi$ built directly from half-angles (mekf_align, fusion.c:1094).

Heading from magnetometer. Rotate m_b by $R(q_{\text{tilt}})$ into the tilt-levelled frame; with horizontal components m_x, m_y there,

$$\psi = \text{atan2}(-m_y, m_x).$$

Full attitude $q = q_\psi \otimes q_{\text{tilt}}$, $q_\psi = [\cos \frac{\psi}{2}, 0, 0, \sin \frac{\psi}{2}]$.

Magnetic reference. On first alignment, `m_ref` (NED, Gauss) is set from the measurement rotated to NED and scaled $\mu\text{T} \rightarrow \text{Gauss}$:

$$m_{\text{ref}} = 0.01 R(q) m_b.$$

As-implemented / audit note. The heading sign is $\text{atan2}(-m_y, m_x)$, **not** $\text{atan2}(+m_y, m_x)$; the levelled field is the true NED field rotated by $-\psi$, so the negative sign recovers vessel yaw. (The `+m_y` form returns $-\psi$, mirrored about north — a historical bug invisible when aligning while pointing north; see the in-code comment at `fusion.c:1104`.) The initial `m_ref` inherits any alignment tilt error in its magnitude/dip, which the quiescence-gated EMA (§4.8) and the WMM invariants (§4.9) later remove; its horizontal **direction** is the heading datum and is never subsequently adapted.

Source: tilt/heading coarse alignment — Farrell 2008 §10; TRIAD lineage Shuster & Oh 1981 (*canonical*).

4.4 Prediction — `mekf_predict()` (`fusion.c:1134`)

Per IMU sample, with measured interval dt (from hardware timestamps, §11; falls back to nominal $1/\text{ODR}$).

Bias-corrected rate: $\omega = s.\text{gyro} - b$.

Quaternion propagation by the exponential map (`q_from_rotvec`, `fusion.c:293`):

$$q \leftarrow q \otimes \exp\left(\frac{1}{2} [0, \omega dt]\right), \quad \exp(\phi) = \left[\cos \frac{|\vartheta|}{2}, \frac{\sin(|\vartheta|/2)}{|\vartheta|} \vartheta \right],$$

$\vartheta = \omega dt$, with the small-angle branch $\delta q \approx [1, \frac{1}{2}\vartheta]$ for $|\vartheta| < 10^{-7}$. Renormalized after.

Covariance propagation $P \leftarrow \Phi P \Phi^\top + Q_d$, first-order discrete transition (`mekf_predict`, `fusion.c:1162`):

$$\Phi = \begin{bmatrix} I_3 - [\omega]_\times dt & -I_3 dt & 0_3 \\ 0_3 & I_3 & 0_3 \\ 0_3 & 0_3 & \varphi I_3 \end{bmatrix}, \quad Q_d = \text{diag}(Q_g \frac{dt}{dt_0} I_3, Q_b \frac{dt}{dt_0} I_3, Q_w I_3),$$

$[\omega]_\times$ the skew-symmetric cross-product matrix; the gyro/bias noise is rescaled by dt/dt_0 ($dt_0 = \text{nominal step}$) so variance grows linearly with the real interval. P is symmetrized after. Convergence flag $\text{tr}(P[0:3]) < \text{conv_thresh}$.

The wave block is discretized exactly, not to first order:

$$\varphi = e^{-dt/\tau}, \quad Q_w = \sigma_g^2 (1 - \varphi^2), \quad \hat{a}_w \leftarrow \varphi \hat{a}_w.$$

This is not fastidiousness. τ is of order 0.5 s while the $|a|$ skip band routinely leaves gaps of the same order, so dt/τ is **not** small on exactly the steps that matter, and a first-order expansion would both over-decay the state and mis-size its noise while bridging a gap. The exact form is stationary for any dt : feed it a step of 10τ and it returns the state to zero with variance σ_g^2 , which is the correct answer — so a scheduling hiccup cannot destabilize the filter. It is also why Q_w is *not* put through the dt/dt_0 rescaling: it is already exact for this dt .

With the state disabled $\varphi = 1$ and $Q_w = 0$, leaving an identity block that touches nothing.

As-implemented. Prediction uses the plain $\Phi P \Phi^\top + Q$ form (not Joseph — the Joseph stabilization is applied on the *updates*, §4.5). Φ is a first-order (zero-hold) approximation of $\exp(F_c dt)$; valid because $\|\omega\|dt \ll 1$ at supported ODRs. The $[\omega]_\times$ sign in $\Phi_{[0:3,0:3]}$ is that of the right-perturbation error convention.

Source: Solà 2017 eq. 259 (integration), eq. 268 (covariance) (*code comment*).

4.5 Generic vector measurement update — `eskf_update()` (`fusion.c:495`)

Measurement of a known NED reference observed in body: predicted h , actual z , isotropic noise R_{noise} , gate `chi2_gate`.

Jacobian (3×9), attitude block only (`dir_jacobian`):

$$H = [[h]_{\times} \mid 0_3 \mid 0_3], \quad [h]_{\times} = \begin{bmatrix} 0 & -h_2 & h_1 \\ h_2 & 0 & -h_0 \\ -h_1 & h_0 & 0 \end{bmatrix}.$$

The **gravity** update uses a different Jacobian when the wave state is enabled — see §4.5.1. The mag channels use this one, unchanged.

Innovation covariance and gain:

$$S = HPH^{\top} + R_{noise}I_3, \quad K = PH^{\top}S^{-1},$$

S^{-1} by Cramer's rule (`m33_inv`, `fusion.c:95`; singular $\Rightarrow$ skip).

The singularity test is relative, not absolute (1.7). S carries physical units, and for the gravity update they are tiny: with $R_a \approx 4.2 \times 10^{-5}$ and the attitude block converged to $\sim 0.4^\circ$, $\det S \approx R_a \lambda^2 \approx 6 \times 10^{-13}$ at a condition number of about 3. The previous absolute test $|\det S| < 10^{-12}$ therefore declared a well-conditioned matrix singular and made `eskf_update` return -1 , **silently dropping 87% of accel updates in the wave benchmark** and pinning attitude uncertainty at whatever value made $\det S \approx 10^{-12}$ rather than letting it converge. `m33_inv` now compares $|\det|$ against $10^{-6} \|A\|_{\max}^3$, which is unit-free, tests rank deficiency (what "singular" means), and still sits well above float's $\approx 1.2 \times 10^{-7}$ epsilon. §4.7 records what this exposed.

Innovation $\nu = z - \hat{z}$ ($\hat{z} = h$ here), with **robust (Huber-style) gating** on the Mahalanobis distance $d^2 = \nu^{\top} S^{-1} \nu$:

$$\begin{cases} \text{reject the update} & d^2 > 9\gamma \\ \nu \leftarrow \nu \sqrt{\gamma/d^2} & \gamma < d^2 \leq 9\gamma \\ \nu \text{ unchanged} & d^2 \leq \gamma \end{cases}$$

$\gamma = \text{chi2_gate}$. The middle branch caps the innovation's influence at the gate distance rather than discarding it.

Correction and injection:

$$\delta x = K \nu, \quad q \leftarrow q \otimes \exp(\delta \theta), \quad b \leftarrow b + \delta b, \quad a_w \leftarrow a_w + \delta a_w,$$

$\delta \theta = \delta x[0:3]$, $\delta b = \delta x[3:6]$, $\delta a_w = \delta x[6:9]$; quaternion renormalized. The a_w injection is unconditional: with the state disabled P 's wave rows are zero, so K 's are too and it adds exactly $0.0f$. The *mag* channels inject it as well — their H has no wave block, but the cross-covariance the accel path builds up means K does, and that is correct Kalman book-keeping.

Covariance — Joseph form, with error-state reset folded in:

$$P \leftarrow [G(I - KH)] P [G(I - KH)]^{\top} + R_{noise} (GK)(GK)^{\top},$$

then symmetrized. (Isotropic $R \Rightarrow K R K^{\top} = R_{noise} K K^{\top}$.)

Error-state reset. Injecting $\delta \theta$ into the quaternion moves the linearisation point, so P is rotated by the reset Jacobian $G = I - \frac{1}{2}[\delta \theta]_{\times}$ (Solà eq. 285). Only the attitude block resets — the gyro-bias and wave-acceleration errors are additive and carry over — so the applied transform is $G_{full} = \text{diag}(G, I_3, I_3)$, implemented by premultiplying the first three rows of K and $(I - KH)$. Cost is ~ 81 multiply-adds against

the Joseph block's ~ 450 . Measured over the 12-seed wave benchmark this improves yaw-only attitude RMS $4.10^\circ \rightarrow 3.57^\circ$ and covariance consistency (NEES $32.4 \rightarrow 28.4$), with 3-D attitude neutral (+2%).

Huber cap. The cap replaces a hard χ^2 reject so that wave-orbital acceleration (which swings gravity *direction* while keeping $|a| \approx g$) deweights rather than starves the filter; only gross outliers ($d^2 > 25\gamma$, GROSS_REJECT_MULT) are rejected. Because S contains P , the cap is naturally inactive during acquisition (large P) and engages only once confident — no explicit convergence gate needed.

Gross-reject threshold (1.7: $9\gamma \rightarrow 25\gamma$). The original 9γ was too tight to be a fault gate. Over the 12-seed benchmark it discarded 26% of 3-D accel updates — routine wave motion, not outliers — which starved the filter, and the resulting drift produced larger innovations that tripped the gate more often still. Widening it (3-D / yaw-only pairs):

reject	3-D att	yaw att	NEES(tr) 3-D/yaw	NIS 3-D/yaw	reject rate
9γ	6.85°	3.57°	28.9 / 21.1	30.7 / 27.3	.262 / .043
16γ	5.45°	2.32°	16.7 / 7.9	22.3 / 25.2	.012 / .000
25γ	5.65°	2.31°	18.3 / 7.8	19.3 / 25.2	.007 / .000
64γ	5.62°	2.31°	18.3 / 7.8	16.7 / 25.2	.000 / .000

Attitude RMS improves 17% (3-D) and 35% (yaw-only), NEES roughly halves, and the worst-draw spread collapses (3-D $13.7^\circ \rightarrow 7.0^\circ$, yaw $9.6^\circ \rightarrow 2.3^\circ$) — the spread §10.8 had attributed to scenario luck. The benefit saturates by 25γ , which keeps a genuine fault gate at $5\times$ the cap radius instead of $3\times$.

Known inconsistency (measured, deliberately retained). The cap scales ν but leaves K — and hence the covariance update — at full confidence, so P contracts as though the measurement had been fully trusted even when the correction was attenuated up to $3\times$. Making this consistent was tried three ways and measured over the 12-seed wave benchmark, where NEES ≈ 1 would indicate a self-consistent filter:

Re-measured after the 25γ change above (the earlier figures were taken while the reject gate was corrupting every run). Rebuild with `-DHUBER_VARIANT=n`:

n	variant	3-D att	yaw att	NEES(tr) 3-D/yaw	NIS 3-D/yaw
0	ν -capping (shipped)	5.65°	2.31°	18.3 / 7.8	19.3 / 25.2
1	$K \leftarrow wK$ (exact Joseph, suboptimal gain)	8.85°	3.09°	39.9 / 12.4	17.0 / 28.5
2	$R \rightarrow R/w$ (IRLS inflation)	7.79°	5.57°	31.5 / 40.5	15.1 / 43.3
3	$R \rightarrow R/w^2$	8.48°	4.74°	35.0 / 27.9	18.2 / 36.3

Every variant is **both** less accurate and less self-consistent — by a wider margin than before. Inflating R is additionally wrong on its own terms because $P \gg R$ here, a regime where $K = P/(P + R)$ is near unity and $R \rightarrow R/w$ barely moves it, losing the bounded per-sample influence the seaway robustness depends on.

The reason these fail is **not** that R_a is mistuned. R_a is over-optimistic (NIS ≈ 20 , §4.7), but retuning it does not rescue the family — they stay worse across the whole sweep. The cap is a *robustness* device, not a statistical one: contracting P by the attenuated gain faithfully records having learned less from that sample, but the consequence is a larger P , hence a larger gain on the **following** samples — which in a seaway carry the same wave contamination. The inconsistency does useful work, holding the gain down exactly when measurements are least trustworthy. `innov_weight` / `innov_reject` on the wire (§8.1) expose how hard the cap is being leaned on; `nis_accel` / `nis_mag` (§8.2) expose whether the model under it is right.

Source: EKF update Kalman 1960; Joseph form Bucy & Joseph 1968; robust innovation capping Huber 1964; Mahalanobis gating Bar-Shalom et al. 2001 (*canonical*). Jacobian sign Solà 2017 §7 (*code comment*).

4.5.1 Wave-aware gravity Jacobian — `wave_jacobian()` When the Gauss–Markov state is enabled, the gravity update no longer predicts h ; it predicts the *contaminated* direction the accelerometer actually sees. With the specific force written in normalized gravity units,

$$v \equiv h - \hat{a}_w, \quad \hat{z} = \frac{v}{\|v\|}, \quad \nu = z - \hat{z}.$$

Perturbing both contributions ($h_{true} = \hat{h} + [\hat{h}]_{\times} \delta\theta$, $a_{true} = \hat{a}_w + \delta a_w$) gives $\delta v = [\hat{h}]_{\times} \delta\theta - \delta a_w$, and differentiating the normalization introduces the **tangent projector** $P_{\hat{z}} = I - \hat{z}\hat{z}^T$:

$$\delta \hat{z} = \frac{P_{\hat{z}}}{\|v\|} \delta v \implies H = \left[\begin{array}{c|c} \frac{1}{\|v\|} P_{\hat{z}} [\hat{h}]_{\times} & 0_3 \\ \hline -\frac{1}{\|v\|} P_{\hat{z}} & \end{array} \right].$$

The projector is load-bearing, not tidiness. It is what makes $\hat{z}^T H = 0$ hold *exactly*, since $P_{\hat{z}} \hat{z} = 0$. That in turn makes $\hat{z}$ an eigenvector of S with eigenvalue R , which is the sole premise of the $\mathbb{E}[d^2] = 2$ derivation in §8.2 — i.e. of the `nis_accel` wire field meaning "1.0 = consistent". Writing the Jacobian the way a first pass naturally would, $[\hat{h}]_{\times}/\|v\|$ and $-I/\|v\|$, leaves $\hat{z}^T H = O(\|\hat{a}_w\|)$; the radial component then leaks into d^2 and NIS is biased. Measured: the unprojected form moves the ground-truth benchmark NIS from 0.99 to 0.76 while leaving every other test in the suite passing (`test_wave_gm_ground_truth`).

At $\hat{a}_w = 0$ this reduces algebraically to `dir_jacobian`: $v = h$ is already a unit vector and $(I - hh^T)[h]_{\times} = [h]_{\times}$ because $h^T[h]_{\times} = 0$.

$\|v\|$ is guarded at 0.5; below that the acceleration is comparable to gravity itself and the linearisation is meaningless, so the update falls back to the plain direction form.

4.6 Scalar heading update — `eskf_update_yaw()` (`fusion.c:758`)

Heading-only correction (a 1-D measurement). Innovation y (rad, wrapped to $\pm\pi$), variance R_{noise} . The Jacobian projects the error onto the NED-down axis expressed in body:

$$H = \left[\begin{array}{c|c} -R[2][:] & 0_3 \end{array} \right] \in \mathbb{R}^{1 \times 6},$$

$R[2][:]$ = third row of $R(q)$. Then

$$S = HPH^T + R_{noise}, \quad K = PH^T/S,$$

with the same Huber policy on $d^2 = y^2/S$ against $\gamma_{\psi} = 6.63$ (χ_1^2 99%), rank-1 Joseph covariance update.

Source: scalar-measurement EKF specialization of §4.5 (*canonical*).

4.7 Accelerometer update — `mekf_update_accel()` (`fusion.c:1374`)

Speed-aided centripetal correction. In a turn the accelerometer senses $a = a_{platform} - g$, and with speed-over-ground v (body $\approx [v, 0, 0]$, leeway neglected) the centripetal term is $\omega \times v = [0, \omega_z v, -\omega_y v]$. When `speed_mps` > 0.1:

$$a_y \leftarrow a_y - \omega_z v, \quad a_z \leftarrow a_z + \omega_y v,$$

$\omega_{y,z}$ bias-corrected. `speed_mps=0` (no GPS) $\Rightarrow$ no-op.

Quiescence EMA (updated every sample, incl. gated), $\tau \approx 2$ s:

$$q_{quiet} \leftarrow q_{quiet} + ((a_g - 1)^2 - q_{quiet}) \frac{dt}{2}, \quad a_g = \|a\|/g.$$

Skip gate. If $a_g \notin [\text{accel_skip_lo}, \text{accel_skip_hi}]$, skip (linear acceleration).

Measurement. Gravity direction $z = -a/\|a\|$; prediction $h = R^T g_{ref} = R[2][:]$ (third row). Effective noise $R_{eff} = R_a \cdot \text{Ra_scale}$. Then `eskf_update(h, z, R_eff, 11.34)` with $\gamma = 11.34$ (χ_3^2 99%). The gravity direction z is stashed as `g_body` (attitude-independent dip reference for §4.8).

Protected axis. h is also passed as `eskf_update`'s `protect` argument, which projects the attitude rows of K onto the plane orthogonal to it:

$$K_{\delta\theta} \leftarrow (I - hh^\top) K_{\delta\theta}.$$

Two properties follow, and they are what the update is entitled to claim:

$$\delta\theta \cdot h = 0, \quad h^\top P^+ h = h^\top P^- h,$$

the second because $K^\top(h, 0, 0) = (h^\top K_{\delta\theta})^\top = 0$ leaves both $(I - KH)^\top$ and $KRK^\top$ inert along h . Joseph form holds for any gain, so nothing else in §4.5 changes.

$H_{\delta\theta}h = 0$ already holds exactly for both Jacobians — the projection subtracts nothing at small angle. It exists for the large-angle regime. Once $\text{tr } P[0:3]$ reaches $O(1)$ rad², which is where an unaided yaw axis ends up, the gain is large enough that a single innovation moves q further than the first-order reset $G = I - \frac{1}{2}[\delta\theta]_\times$ tracks h ; the misalignment leaks the unobservable variance into the observable subspace, the next update annihilates it, and the filter reports a collapsed covariance and a heading correction gravity cannot justify.

Attitude rows only. A gyro-bias component along h is equally unobservable from this measurement at this instant, but h rotates in the body frame as the platform moves, so that component becomes observable a moment later. A yaw error does not: Φ rotates $\delta\theta$ with the body and h rotates with the body, so $\delta\theta \parallel h$ stays parallel to h for all time. Projecting rows 0–2 is therefore correct and projecting rows 3–5 would not be. The magnetometer passes `protect = NULL` (§4.8): it is the measurement that does carry heading.

As-implemented. `Ra_scale` is set to 4 by the fusion thread while engine vibration is detected (§9) — vibration is high-frequency, near-zero-mean, so dewatering (not gating) is correct. Wave *direction* disturbance is handled by the Huber cap (§4.5), **not** by magnitude-based dewatering, which benchmarking showed rectifies wave-phase-correlated error into an attitude/bias offset. The centripetal model assumes zero leeway and no vertical velocity.

Health-EMA gain (1.7 fix). `innov_weight`, `innov_reject` and `nis_accel` are fed only by samples that clear the skip gate. In a seaway that gate discards 85–95% of samples, so a per-update gain sized from the IMU ODR (`gate_alpha = 1/(\tau f_{odr})`) stretched the intended $\tau = 30$ s to ten minutes or more, and the metrics reported their seed values rather than the data. The accel path now derives the gain from elapsed time instead:

$$\alpha = 1 - \exp(-\Delta t/\tau),$$

Δt accumulated over *every* sample, accepted or skipped. This is exact at any feed rate and reduces to $\Delta t/\tau$ when updates are frequent.

What R_a can and cannot do. R_a derives from the datasheet noise floor, and the filter is measurably over-confident about it: mean NIS over the wave benchmark is 19 (3-D) / 25 (yaw-only) where a consistent model reads 1. The natural remedy — raise `mekf_accel_noise` until $\text{NIS} \approx 1$ — does not work. Sweeping it (12 seeds, both mag modes):

N_a	3-D att	yaw att	NEES(tr) 3-D/yaw	NIS 3-D/yaw
0.0022 (default)	5.65°	2.31°	18.3 / 7.8	19.3 / 25.2
0.004	7.93°	11.13°	114 / 304	33.7 / 39.6
0.008	7.57°	11.09°	102 / 285	10.6 / 13.5
0.015	6.98°	10.29°	81 / 241	3.2 / 4.2
0.030	5.74°	8.58°	50 / 156	0.82 / 1.03
0.050	5.07°	7.01°	34 / 87	0.32 / 0.34
0.120	4.72°	4.33°	17 / 17	0.06 / 0.05
0.300	4.47°	2.48°	6.4 / 2.2	0.01 / 0.01

Three things to read off it:

1. **NIS ≈ 1 costs accuracy.** $N_a = 0.03$ makes the innovations statistically consistent, but the marine (yaw-only) default degrades $2.31^\circ \rightarrow 8.58^\circ$ and its NEES rises $7.8 \rightarrow 156$. Weakening the gravity correction makes the filter lean on gyro integration, so the *state* error grows even as the *innovations* start to look honest.
2. **The two consistency criteria disagree.** NIS falls monotonically with N_a while NEES(trace) has its minimum near the default. No single scalar satisfies both.
3. **The inconsistency is structural, not a scale error.** NEES(strict) for 3-D stays pinned in the 44–64 band across the entire sweep — four orders of magnitude of R_a — so P 's *shape* is wrong, and no isotropic scalar can fix that.

The cause is that the seaway residual is wave-orbital: correlated with wave phase, with a measured correlation time of order a second (`imud-cal fit-ra`), not white. A white isotropic R cannot describe a coloured disturbance; making its *variance* right necessarily gets its *spectrum* wrong. The real fix is to model the correlation, which is what §4.7.1 does.

The default is therefore unchanged, and is additionally a sharp local optimum: $N_a \in [0.003, 0.03]$ is markedly **worse** than either side of it, so hand- tuning it upward "a little" lands in the worst available region.

Caveat on the table above. These rows were measured before the `m33_inv` singularity bug of §4.5 was found, i.e. through a filter that was discarding 87% of its accel updates. The *conclusions* survive — no scalar R_a fixes a coloured disturbance, and the 44–64 NEES(strict) band really was structural — but the absolute numbers are not reproducible on current code and the table is kept as the historical record of why §10.5 was undertaken.

4.7.1 The Gauss–Markov wave state — outcome Fixing `m33_inv` removed an accidental 87% decimation of accel updates. That decimation had been doing real work: it crudely decorrelated the wave-contaminated samples, which is why the filter looked as good as it did. With every sample fed to a white R , the filter believes it has 833 independent gravity measurements per second, P collapses, and it diverges. Modelling the disturbance as a first-order Gauss–Markov process (§4.1.1) makes repeated correlated samples correctly stop adding information. Over the 12-seed wave benchmark:

	3-D att	3-D hdg	yaw att	yaw hdg	NEES(tr) 3-D/yaw	NEES(st) 3-D/yaw	NIS 3-
old baseline (bug present)	5.653°	3.065°	2.309°	1.961°	18.3 / 7.8	57 / 38	19.3 /
<code>m33_inv</code> fixed, no wave state	9.488°	7.255°	11.424°	8.801°	259 / 252	933 / 1086	56.5 /
+ wave state (shipped)	4.452°	0.828°	2.308°	1.016°	3.47 / 0.99	335 / 10.1	1.01 /

(For 3-D attitude and NEES(strict), §4.8.1 carries the authoritative figures: it measures them with the benchmark's alignment corrected. The wave-state conclusion here rests on the yaw-only default and on the NIS column.)

Against the old baseline: 3-D attitude -21% , 3-D heading -73% , yaw heading -48% , yaw attitude a dead heat, NIS from 19–25 to ≈ 1 , NEES(trace) from 18.3/7.8 to 3.47/0.99, and both the Huber cap and the gross-reject gate go completely idle (weight 1.000, reject 0.0000) — the filter no longer needs robustness machinery to survive an ordinary seaway.

Tuning, and why it is not curve-fitting. σ and τ were chosen on a *broadband* scenario (`SCEN_GM` in `test_fusion.c`) whose disturbance is a genuine Gauss–Markov process of known σ and τ , because a single tone has no correlation time and fitting τ to one is fitting the benchmark. The tone is the held-out validation. Three independent lines agree on $\sigma \approx 0.8 \text{ m} \cdot \text{s}^{-2}$:

- it is the broadband scenario's ground truth, at which the filter reports $\text{NIS} = 0.99$ (`test_wave_gm_ground_truth`);
- it is exactly the tone scenario's true per-axis RMS ($1.2/\sqrt{2}$);
- `imud-cal fit-ra` independently recovers $0.67 \text{ m} \cdot \text{s}^{-2}$ from a capture of that tone, seen only through a file, a replayed filter and a 67%-decimating skip band.

$\tau = 0.5$ s sits inside the range fit-ra measures. Neither knob is a free parameter. The full grid is reproducible with

```
rm -f test_fusion && make test_fusion CFLAGS="-D_GNU_SOURCE -O2 -Wall \
-Wextra -std=c11 -pthread -linclude -DBENCH_SWEEP_WAVE" && ./test_fusion
```

The degeneracy ridge. $\delta\theta$ and δa_w are separated only by their dynamics, so an over-large τ lets the wave state absorb real tilt. The sweep shows exactly that: at $\sigma = 0.9$ the yaw-only attitude RMS runs $2.17^\circ / 2.29^\circ / 2.40^\circ / 2.72^\circ$ for $\tau = 0.3 / 0.5 / 0.6 / 0.8$ s, and by $\tau = 2$ s it has collapsed to 7.14° . Short τ is the safe side; the shipped 0.5 s sits at the knee. A visible symptom of straying too far is `bias_z` drifting, which the benchmark bounds independently.

The column that looked worse: 3-D NEES(strict), 57 $\rightarrow$ 335. Two things were going on, and the first attribution of this to "the swing-circle mag calibration is structurally 2-D" was **wrong** — the benchmark synthesises its magnetometer data from the true attitude with nothing but white noise, so no calibration defect can be what it measures. See §4.8.1: it is the magnetic reference's DIP error, and $\sim 96\%$ of it was the benchmark aligning from a single instantaneous sample where the daemon averages a window.

4.7.2 The filter across the whole rate ladder Every figure above, and every figure elsewhere in this document, was measured at one point: 833 Hz IMU with a ~ 104 Hz magnetometer. The drivers advertise **29 IMU rates from 12 Hz to 32 kHz and 13 magnetometer rates from 1 Hz to 1 kHz**, and each of those 377 pairings is a configuration `imud.conf` will accept. This section is what the other 376 do.

Two instruments, because the questions are different sizes. `test_rate_derivations` walks all 377 pairings on every build, asserting the derived tuning is correct and the filter stays numerically sane — it deliberately asserts nothing about accuracy, since $R_a \propto f_{odr}$ retunes the filter at every rung and the 833 Hz bounds are meaningless elsewhere. Accuracy comes from an opt-in sweep along two axes:

```
rm -f test_fusion && make test_fusion CFLAGS="-D_GNU_SOURCE -O2 -Wall \
-Wextra -std=c11 -pthread -linclude -DBENCH_SWEEP_ODR" && ./test_fusion
```

Accuracy is very nearly rate-independent. 3-D attitude RMS over the IMU axis, mag pinned at 100 Hz:

f_s (Hz)	12	26	52	104	208	416	833	1660	3332	6664	8000	16000	3
3-D att RMS	1.210°	1.163°	1.156°	1.181°	1.157°	1.176°	1.178°	1.192°	1.210°	1.224°	1.252°	1.280°	1
yaw att RMS	7.673°	2.370°	1.980°	2.036°	2.083°	2.105°	2.185°	2.203°	2.226°	2.241°	2.232°	2.241°	2
NIS _a (3-D)	15.66	7.40	3.84	2.11	1.26	0.82	0.63	0.53	0.47	0.42	0.42	0.35	0

$\pm 6\%$ about the midpoint across a $2667\times$ range in 3-D — 1.156° at 52 Hz to 1.299° at 32 kHz. A low-power board at 104 Hz gives up essentially nothing, which was the question that prompted the work.

The top two rungs arrived later, with the ODR-coverage work that added icm42688p's 16 kHz and 32 kHz. They did not change the conclusion, and they did not reveal a new mechanism either: 3-D attitude RMS has climbed steadily above the 833 Hz default for the whole length of this table, and it simply keeps climbing. Through 8 kHz the spread was $\pm 4\%$; the two new rungs supply the rest. NIS_a keeps falling too, to 0.23 at 32 kHz — nearly $3\times$ more under-confident than at the default, on the same trend the next paragraph describes rather than a break in it.

Worth separating from a related finding, because they are *not* the same effect. `test_heave_across_rates` measures float32 accumulation error in the heave integrator turning upward above 6664 Hz (0.112% there, 0.247% at 32 kHz; see the comment on that test). The MEKF's attitude degradation here begins at 833 Hz, well below that inflection, and is a property of the tuning across the ladder rather than of float32 accumulation. Two different curves that happen to both bend upward at the top.

But NIS_a is not flat, and crosses 1.0 near 150–200 Hz. The shipped measurement model is therefore most self-consistent around 200 Hz; at the 833 Hz default the filter is mildly *under*-confident, and below ~ 100 Hz increasingly over-confident. Accuracy does not track this at all.

And that is not a tuning choice. Re-running the (σ, τ) grid at 104 Hz: $\tau = 0.5$ s is still at the knee — the eight-fold drop in samples per correlation time, 416 to 52, does not move it — but *no point in the grid* brings NIS_a near 1. The best is ≈ 1.69 at $\sigma = 1.8$, twice the shipped value and at a real accuracy cost. The decisive evidence is **SCEN_GM**, the one configuration whose right answer is known in advance: with the knobs set to the truth it reports $\text{NIS}_a = 0.87$ at 833 Hz and **≈ 3.4 at 104 Hz**. Whatever this is, it is structural, and it is the open question this section leaves behind.

It is *not* the harness. The benchmark injects per-sample sigmas while the filter models a density $\times$ bandwidth, so the scenario scales its sensor draws by $\sqrt{f_s/833}$ (and the magnetometer independently by $\sqrt{f_{mag}/100}$). Cross-checked, R_a over-estimates the injected per-sample variance by $4.480\times$ at 833 Hz and $4.480\times$ at 104 Hz — identical to four figures.

The floor is at 12 Hz, only in yaw-only mode, and it is the gate.

f_s (Hz)	12	14	16	20	25	26	40	52
yaw att RMS	7.673°	3.949°	3.404°	2.838°	2.408°	2.370°	2.043°	1.980°
NEES(strict)	402	102	72.9	46.9	30.0	28.4	15.3	12.1
reject	.1051	.0000	.0000	.0000	.0000	.0000	.0000	.0000

Degradation from 52 Hz down to 14 Hz is smooth; 12 Hz is a cliff. The mechanism is visible in the last row — the gross-outlier gate goes from idle to rejecting 10.5% of updates, which is the rejection-feedback regime §4.7 describes: lost corrections cause drift, drift enlarges innovations, and those trip the gate more often still. 3-D mode at 12 Hz is unaffected (1.210°).

Note also that yaw-only is *best* around 40–52 Hz (1.980°), not at the 833 Hz default (2.185°).

This does not test the $\Phi = I + F_c \Delta t$ linearisation, and the claim in §4.4 that $\|\omega\|\Delta t \ll 1$ at supported ODRs remains unmeasured. The wave scenario's own peak rate is 18.8 °/s, so $\|\omega\|\Delta t = 0.027$ rad even at 12 Hz — nowhere near the limit. At the 2000 °/s full scale the config permits it would be 2.9 rad at 12 Hz. Probing that needs a high-rate-rotation scenario that does not exist.

Two further results worth having. The Gauss–Markov wave state is load-bearing at *every* rate, not a fix specific to 833 Hz: disabled, attitude RMS runs **5.0–11.4°** across the whole ladder — against 1.16–1.30° in 3-D with it enabled — and the gross-outlier gate goes from idle to rejecting **17–29%** of updates at 416 Hz and below. Both ranges are from the re-run that added the 16 kHz and 32 kHz rungs, and both are wider than the 8–11° and 12–27% previously recorded here; the mag-axis figures in the next paragraph reproduce to the digit against the same run, so this is a stale entry rather than a non-deterministic bench. Its *direction* also needs qualifying: the wave state mattering more as the rate falls is true of the rejection rate, not of accuracy. By RMS ratio it is worth least at 52 Hz (4.4 $\times$) and most around the 833 Hz default (8.6 $\times$), tapering to 7.8 $\times$ at 32 kHz. And the magnetometer ladder is benign for accuracy — 3-D attitude is flat at 1.171–1.223° from a 1 Hz mag to a 1 kHz one — while being hostile to the NIS instrument: NIS_m runs 9.84 at 1 Hz against 0.46 at 1 kHz, because $R_m = N_m^2 f_{mag}$ shrinks with the rate while the attitude-error component of the innovation does not. NEES(strict) correspondingly *improves* at low mag rates (0.59 at 1 Hz against 5.74 at 100 Hz), which is independent corroboration of the dip-error diagnosis in §4.8.1: fewer 3-D mag updates inject less alignment dip bias into roll and pitch.

The gyro pad. `mekf_gyro_noise = 0.007` is a $\sim 58\times$ pad standing in for wave-induced angular dynamics. If that were all it covered it should be rate-invariant, since $Q_g = N_g^2 \Delta t$ already delivers the same rad²/second at any rate. Measured, the RMS-optimal N_g is **0.002 at 833 Hz and 0.004 at 104 Hz** — eight times the Δt , twice the pad, close to the $\sqrt{\Delta t}$ that $N_g^2 \Delta t$ implies for an error growing linearly in wall-clock time. That supports intra-sample rotation nonlinearity as (part of) what the pad absorbs.

It does **not** justify changing the default, for the reason §4.7 already documents for R_a : at 833 Hz, $N_g = 0.002$ gives the best attitude RMS in the grid (1.074° against 1.178° shipped) *and* the worst NEES(strict) near it (18.5 against 5.74). No scalar satisfies both. Worth recording that the shipped 0.007 is essentially optimal

for yaw-only (2.185° against a best of 2.178°) while ~9% off optimal for 3-D — a deliberate compromise favouring the marine default, not an arbitrary number. Reproduce with `-DBENCH_SWEEP_NG`, composable with `-DBENCH_ODR_HZ=<rate>`.

4.8 Magnetometer update — `mekf_update_mag()` (`fusion.c:1438`)

Measurement in Gauss ($m = 0.01 m_{\mu T}$). Predicted body field $h_{raw} = R^\top m_{ref}$, magnitude $|h_{raw}|$.

Normalization. Both predicted and measured vectors are unit-normalized before the update; the mag noise is rescaled $R_{m,n} = R_m/|h_{raw}|^2$ to preserve the optimal gain. (Rationale: in Gauss² units $\det S \approx R_m^3 \approx 4 \times 10^{-15}$ falls below the inverse's singularity threshold once P_{att} is small; normalizing makes S dimensionless and well-conditioned.) Residual $\nu = z - h$, $\text{res_sq} = \|\nu\|^2$.

Compass-health metrics — computed **before** the gates (§8).

Gates. Magnitude ratio $r = |m|/|h_{raw}|$ must lie in $[0.5, 2]$; direction $\text{res_sq} < 4$ ($\leq 90^\circ$ correction). The tight anomaly gate is below, after the heading innovation it needs.

Reference adaptation (m_ref EMA), magnitude + dip only. When `mref_alpha`>0, `g_body_valid`, $\text{res_sq} < 0.1$, and $q_{quiet} < 2 \times 10^{-4}$ (platform calm), adapt the field magnitude and dip from **attitude-independent invariants**: $\sin(\text{dip}) = (m \cdot g_{body})/|m|$,

$$m_{ref}^h \leftarrow m_{ref}^h (1 + \alpha_{mref} (|m| \cos \text{dip} / m_{ref}^h - 1)), \quad m_{ref,z} \leftarrow m_{ref,z} + \alpha_{mref} (|m| \sin \text{dip} - m_{ref,z}).$$

The horizontal **direction** is never touched.

Heading innovation. Rotate m to NED and take the angle between the measured and reference field in the horizontal plane,

$$y = \text{atan2}(m_y^{NED}, m_x^{NED}) - \text{atan2}(m_{ref,y}, m_{ref,x}) \text{ (wrapped)}.$$

A near-vertical field ($m^h < 0.2|h_{raw}|$, measured or reference) carries no heading information; y is undefined there and the heading-only update is skipped.

Tight anomaly gate (converged only). The squared residual chord in Gauss², taken in the plane the update actually fuses:

$$\begin{cases} 2(m_{ref}^h)^2(1 - \cos y) > \text{mag_reject_sq} & \text{mag_yaw_only} \\ \text{res_sq} > \text{mag_reject_sq}/|h_{raw}|^2 & \text{otherwise} \end{cases} \Rightarrow \text{skip}.$$

Both are the one test, $2|B|^2(1 - \cos \theta)$ between measurement and prediction, restricted to the fused channel. The 3-vector form counts the dip, which a structurally 2-D swing calibration (§3.3, §12.3) cannot measure, so it stays past the threshold for a healthy compass — applying it to the heading-only path rejects nearly every sample from the moment the filter converges.

Update. If `mag_yaw_only` (marine default): noise $R_\psi = R_m/(m_{ref}^h)^2$, and call the scalar update §4.6 with y . Otherwise call the full vector update §4.5 with $(h, z, R, 11.34)$, where R carries the anisotropic dip term of §4.8.1.

4.8.1 The dip-reference error, and the anisotropic R In 3-D vector mode the field's **dip** constrains roll and pitch. That is the mode's whole value over `mag_yaw_only`, and also its exposure: the dip of m_{ref} is the one part of the reference the filter cannot establish accurately on its own.

m_{ref} is fixed once, at alignment, from the tilt estimate available at that moment (§4.3). In a seaway that tilt is wrong, and the error is baked in permanently — in 3-D mode it becomes a **constant roll/pitch bias** that P has no term for, which is what a large NEES(strict) is reporting. Measured over the 12-seed benchmark — these are the authoritative 3-D figures:

3-D mode	dip error	att RMS	hdg RMS	NEES(tr)	NEES(st)
align from 1 sample	-4.38°	4.452°	0.828°	3.47	335
align over the window	$+0.86^\circ$	1.204°	0.745°	0.21	12.8
+ dip reference exact (WMM)	0.00°	0.841°	0.737°	0.11	0.22

The per-axis decomposition is unambiguous: at the top row the mean attitude error is $[-3.85^\circ, -2.00^\circ, -0.28^\circ]$ against an RMS of $[3.87^\circ, 2.04^\circ, 0.87^\circ]$ — almost pure bias, not spread.

It cannot be healed in run. The `m_ref` EMA above exists for exactly this and is gated on $q_{quiet} < 2 \times 10^{-4}$, which a seaway never reaches ($q_{quiet} \approx 5 \times 10^{-3}$). That gate is correct, not an oversight. Raising it looks like a fix over 120 s and is refuted over 30 minutes:

3-D, 1800 s window	dip error	$ m_{ref}^h $ error	att RMS	NEES(st)
gate 2×10^{-4} (shipped)	$+0.862^\circ$	-5.04%	1.151°	12.84
gate 2×10^{-2}	-1.460°	$+4.24\%$	1.725°	42.10
gate off	-1.469°	$+4.27\%$	1.729°	42.31

The reference sails past truth and keeps going, because the samples clearing the $|a|$ band are wave-phase correlated: learning from that subset walks m_{ref} away rather than toward. (De-contaminating g_{body} with the Gauss–Markov $\hat{a}_w$ was also tried, and is worse — $\hat{a}_w$ has itself absorbed some tilt, so adding it back re-injects a correlated error.)

So the dip error is **not observable from seaway data**. It is removed at the source by WMM invariants (`mekf_set_mref_invariants`), or it is admitted into P .

The anisotropic term. A dip error $d\delta$ rotates m_{ref} about $\hat{a}_{NED} = \hat{h}_{hor} \times \hat{e}_D$, so in body frame it perturbs the normalised prediction along one unit tangent direction:

$$\hat{a}_{body} = R^\top \hat{a}_{NED}, \quad u = \hat{a}_{body} \times \hat{h}, \quad \|u\| = 1, \quad u^\top \hat{h} = 0,$$

(the last two because $\hat{m}_{ref}$ lies in the plane spanned by $\hat{h}_{hor}$ and $\hat{e}_D$, hence $\hat{m}_{ref} \perp \hat{a}_{NED}$). The uncertainty is therefore exactly rank-1:

$$R = R_{m,n} I_3 + \sigma_{dip}^2 uu^\top,$$

with $\sigma_{dip} = \text{mekf_mag_dip_sigma_deg}$ in radians (a dip error of δ displaces the predicted unit direction by δ , so the units match directly). Default 1.0° , which is the measured $+0.86^\circ$ residual rounded up — **not** a value fitted to the metric.

Why rank-1 and not just a larger R_m . Isotropic inflation reaches the same NEES: $R_m \times 64$ gives NEES(strict) $12.83 \rightarrow 1.57$ with attitude and heading both marginally better. But it deweights the heading-carrying components too, and `nis_mag` falls to **0.01** — the wire's magnetometer-health instrument stops being able to report a fault. The rank-1 form leaves those components alone. Sweep, 3-D mode with a window-aligned reference:

σ_{dip}	att RMS	hdg RMS	NEES(st)	nis_mag
0°	1.204°	0.745°	12.83	0.52
0.5°	1.186°	0.723°	9.06	0.39
1.0° (shipped)	1.178°	0.714°	5.74	0.31
2.0°	1.177°	0.715°	3.14	0.28
3.0°	1.179°	0.720°	2.15	0.27

dof = 2 survives. Because u is tangent, $R\hat{h} = R_{m,n}\hat{h}$, so $\hat{h}$ remains an eigenvector of $S = HPH^\top + R$ with eigenvalue $R_{m,n}$; combined with $H^\top\hat{h} = 0$ the §8.2 derivation $E[d^2] = 3 - R_{m,n}\hat{h}^\top S^{-1}\hat{h} = 2$ is unchanged. An anisotropy with any radial component would break it silently — and would also stop the term working at all, since the radial direction of a normalised measurement carries no information (measured: a contaminated u widens P by 1% instead of 9.5% and absorbs 53% of the pull instead of 93%).

Note `nis_mag` settling near **0.5** once σ_{dip} dominates is the correct reading, not a regression: one of the two tangent degrees of freedom is deliberately deweighted, so a consistent filter reads about half.

What this does not do. It does not reach NEES(strict) = 1, and cannot. The dip error is a *bias*; a covariance term can only partly stand in for one, and the accelerometer legitimately keeps P tight in roll and pitch. Driving the number to 1 would need $\sigma_{dip} \approx 4^\circ$, i.e. inventing uncertainty to satisfy a statistic. For an install that cares, the answer is a position source.

As-implemented / audit note. Adapting **only** magnitude and dip is deliberate: the horizontal direction *is* the heading reference, so learning it from the filter's own attitude would feed the heading estimate back into its own datum and let both random-walk together (gauge drift). The adaptation targets are body-frame invariants (total magnitude; dip from the mag–gravity angle), so the filter's attitude never enters. Heading-only fusion is the default because the soft-iron matrix is fitted from a 2-D swing (§3.3, §12.3) and its dip channel is least trustworthy.

Source: heading-only / tilt-compensated magnetometer fusion — standard marine AHRS practice, Farrell 2008 (*canonical*); gauge-feedback avoidance is imud-specific design (see the in-code rationale in `mekf_update_mag, fusion.c:1550`).

4.9 WMM reference invariants — `mekf_set_mref_invariants()` (`fusion.c:1057`)

Given WMM horizontal magnitude H and vertical Z (Gauss) at a known position (from §13), rescale `m_ref` to those invariants while preserving horizontal direction: with $m_h = \sqrt{m_{ref,x}^2 + m_{ref,y}^2}$ and $s = H/m_h$,

$$m_{ref,x} \leftarrow s m_{ref,x}, \quad m_{ref,y} \leftarrow s m_{ref,y}, \quad m_{ref,z} \leftarrow Z.$$

No-op before alignment or if $H \leq 0$.

Source: imud-specific (WMM supplies the invariants of §13).

4.10 State extraction — `mekf_get_state()` (`fusion.c:2006`)

Euler angles from $R(q)$ (NED 3-2-1 aerospace):

$$\theta = \arcsin(-R[2][0]), \quad \phi = \operatorname{atan2}(R[2][1], R[2][2]), \quad \psi = \operatorname{atan2}(R[1][0], R[0][0]).$$

Magnetic heading ψ wrapped to $[0, 360^\circ)$. Attitude covariance (`cov[9]`) = $P[0:3, 0:3]$; bias variance = $\operatorname{diag} P[3:6]$; `quiescence` = q_{quiet} . `rate_of_turn` is left 0 here and filled by the fusion thread (§5).

Source: quaternion→Euler, Diebel 2006 (*canonical*).

4.11 Reconfigure — `mekf_reconfigure()` (`fusion.c:1936`)

Recomputes Q_g, Q_b, R_a, R_m , the skip band, `mag_reject_sq`, `mag_yaw_only`, `mref_alpha`, `conv_thresh`, and the gate-health EMA rate from a new config on hot-reload; q, P, b, dt and the runtime scalars (`Ra_scale`, `acc_quiet_ema`, the health EMAs) are untouched, so the filter keeps running. Both this and `mekf_init()` derive their tuning through one shared helper (`mekf_derive_tuning, fusion.c:864`), so no value can be updated in one path and forgotten in the other — `mref_alpha` previously was, leaving the `m_ref` EMA at a stale rate after any `mag_odr_hz` change.

5. Euler rates and rate of turn

Computed in the fusion thread (`fusion_thread`, `imu.c:1304–1351`) from the bias-corrected body rate $\omega = s.\text{gyro} - b$ and the current Euler angles, using the inverse of the 3-2-1 kinematic relation.

Roll rate (for sea-state, §7), with a near-gimbal-lock fallback:

$$\dot{\phi} = \omega_x + \tan \theta (\omega_y \sin \phi + \omega_z \cos \phi), \quad \dot{\phi} = \omega_x \text{ if } |\cos \theta| \leq 0.2.$$

Pitch rate (no singularity):

$$\dot{\theta} = \omega_y \cos \phi - \omega_z \sin \phi.$$

Rate of turn (heading rate, output in $\text{deg} \cdot \text{min}^{-1}$):

$$\dot{\psi} = \frac{\omega_y \sin \phi + \omega_z \cos \phi}{\cos \theta}, \quad \dot{\psi} = \omega_z \text{ if } |\cos \theta| \leq 0.2, \quad \text{ROT} = \dot{\psi} \cdot \frac{180}{\pi} \cdot 60.$$

As-implemented. ROT is the *Euler yaw rate*, not raw body-Z gyro: at 20° roll the body-Z rate under-reads heading rate and couples in pitch rate, which autopilots notice. Near $\pm 90^\circ$ pitch ($|\cos \theta| \leq 0.2$) the code falls back to the body-axis rate to avoid division blow-up.

Source: 3-2-1 Euler kinematics, Titterton & Weston 2004 §3.6; Diebel 2006 (*canonical*).

6. Heave estimator — `heave_update()` (`fusion.c:1789`)

Vertical displacement from vertical acceleration, by leaky double integration followed by a true first-order high-pass.

Vertical acceleration (NED down, zero at rest):

$$a_D = (R(q) a_b)_D + g = R[2][:] \cdot a_b + g.$$

Leaky double integration, leak $\ell = dt/\tau$:

$$v \leftarrow (v + a_D dt)(1 - \ell), \quad d \leftarrow (d + v dt)(1 - \ell).$$

Output high-pass (exact zero at DC), $\alpha_{hp} = \tau/(\tau + dt)$:

$$y \leftarrow \alpha_{hp} (y + d - d_{prev}), \quad d_{prev} \leftarrow d,$$

heave (positive **up**) = $-y$. `heave_rate` output = $-v$. Settled after $\geq 10\tau$ (**settled** flag).

As-implemented / audit note. Two leaky integrators bound drift but retain a DC gain of $\sim \tau^2$ (≈ 144 m per $\text{m} \cdot \text{s}^{-2}$ at the default $\tau = 12$ s); any slow residual (accel bias, small tilt error, start-up transient) would appear at metre scale. The output high-pass has a **structural** zero at DC, killing those residuals rather than estimating them. Passband amplitude error $\approx 2\%$ at 8 s, $\approx 4\%$ at 12 s for $\tau = 12$ s (band-limited: very long-period swell is attenuated). $\tau = 0$ disables. The estimate is a filtered kinematic quantity, not a statistically-optimal one.

Source: leaky integrator + first-order high-pass are standard discrete-time filter forms, Oppenheim & Schaffer 2010; the double-integration- plus-high-pass heave technique is long-standing in wave-buoy practice (Datawell / Longuet-Higgins tradition) (*canonical*).

7. Sea-state statistics — seastate_* (fusion.c:1847–1934)

Windowed spectral moments over the heave, roll, and pitch oscillations via exponentially-weighted mean/variance pairs — no FFT, no sample storage.

EW mean/variance recursion (ew_stat, fusion.c:1865), $\alpha = dt/\tau$:

$$\mu \leftarrow \mu + \alpha d, \quad \sigma^2 \leftarrow \sigma^2 + \alpha((1 - \alpha)d^2 - \sigma^2), \quad d = x - \mu.$$

Applied to six signals: heave, heave-rate, roll, roll-rate, pitch, pitch-rate. Fed **only while heave is settled** (seastate_update, fusion.c:1872).

Outputs (from the variances; $m_0 = \text{var}(x)$, $m_2 = \text{var}(\dot{x})$):

$$\begin{aligned} H_s &= 4\sqrt{m_0^{heave}}, & T_z &= 2\pi\sqrt{\frac{m_0^{heave}}{m_2^{heave}}}, & T_{roll} &= 2\pi\sqrt{\frac{m_0^{roll}}{m_2^{roll}}}, \\ A_{roll} &= 2\sqrt{m_0^{roll}}, & A_{pitch} &= 2\sqrt{m_0^{pitch}}, & T_{pitch} &= 2\pi\sqrt{\frac{m_0^{pitch}}{m_2^{pitch}}}. \end{aligned}$$

Guards: all return 0 until settled ($\geq 2\tau$); periods additionally return 0 below the oscillation floors $\sigma(\text{heave}) < 2 \text{ cm}$ (SEASTATE_MIN_HEAVE_SIG) or $\sigma(\text{angle}) < 0.3^\circ$ (SEASTATE_MIN_ANGLE_SIG), or non-positive rate variance.

As-implemented / audit note. $H_s = 4\sqrt{m_0}$ is the significant wave height for a narrow-band Gaussian sea (Anguet-Higgins). The period identity $T = 2\pi\sqrt{m_0/m_2}$ is Rice's mean up-crossing period from the zeroth and second spectral moments — **exact for a pure sine** ($\text{var} = A^2/2$, $\text{var}(\dot{x}) = A^2\omega^2/2$, $\text{ratio} = 1/\omega^2$) and an estimator for a real seaway. Amplitudes are reported as **significant single amplitudes** (2σ , seakeeping convention); wave height is the significant **double** amplitude (4σ). The EW variance is biased toward the window's recent history and toward the EW mean (so a steady heel / trim is removed from roll/pitch, appearing in μ , not σ). The default $\tau = 120$ s is far shorter than the 10–20 min oceanographic record; it favours a responsive live display. Roll/pitch **rates** are Euler rates (§5), not body rates.

Source: $H_s = 4\sqrt{m_0}$ Longuet-Higgins 1952; spectral-moment up-crossing period Rice 1944–45; engineering treatment Tucker & Pitt 2001; roll-period/seakeeping convention Lloyd 1989 (*canonical*). EW variance recursion, West 1979 / Finch 2009 (*canonical*).

8. Compass-health diagnostics — mekf_update_mag() (fusion.c:1438)

Two EMAs ($\alpha = 1/3000$, $\tau \approx 30$ s at 100 Hz mag ODR), updated **before** the rejection gates so that gated-out anomalies still register:

Field-magnitude anomaly (attitude-independent): $\text{mag_anom} \leftarrow \text{mag_anom} + \alpha \left(\frac{1}{\text{vert}}, \frac{|\text{mag} - \text{h_raw}|}{\text{vert}} \right) \left(\frac{|\text{h_raw}|}{\text{vert}} \right)$

- \text{mag_anom}\right). \$\$

Heading residual (mode-independent; $m^{NED} = R(q) m$):

$$y = \text{atan2}(m_u^{NED}, m_x^{NED}) - \text{atan2}(m_{ref,u}, m_{ref,x}) \text{ (wrapped to } \pm\pi), \quad \text{mag_resid} \leftarrow \text{mag_resid} + \alpha(|y| - \text{mag_resid}),$$

gated on both horizontal magnitudes exceeding $0.2|h_{r_{gw}}|$.

As-implemented. Diagnostics only — they never modify the filter. Fed from pre-gate innovations because the rejected samples are precisely the anomalies these metrics exist to surface. The 30 s time constant scales inversely with a non-standard mag rate (acceptable for a health indicator).

8.1 Update-gate health — `gate_health()` (`fusion.c:159`)

same statistic offline from a capture, through the filter's own `mekf_accel_probe()`, so the live number can be checked against real water.

Source: NIS consistency test Bar-Shalom et al. 2001 §5.4 (*canonical*).

9. Engine-vibration detector — `ism_reader_thread()` (`imu.c`)

EMA of squared specific-force deviation from 1 g, stepped once per sample:

$$e \leftarrow e + \alpha((\|a\| - g)^2 - e), \quad \alpha = \frac{1}{\tau f_{ODR}}, \quad \tau = 1 \text{ s.}$$

α is derived from the **resolved sample rate**, not fixed. Because the EMA advances once per sample but samples arrive in FIFO bursts, a constant α made the effective time constant a function of burst depth (a deeper burst = more steps per read = a faster EMA) rather than the intended 1 s.

Assertion uses **hysteresis** rather than a bare threshold:

$$\text{engine_on} = \begin{cases} \text{true} & e > \text{engine_vibration_g2} \\ \text{false} & e < 0.7 \text{ engine_vibration_g2} \\ \text{unchanged} & \text{otherwise} \end{cases}$$

A single threshold chattered at the boundary, and each toggle steps `Ra_scale`, rewrites the skip band, emits a log line, and flips `FLAG_ENGINE_ON` on the wire — all externally visible. `engine_on` raises the accel noise $\times 4$ and widens the skip band (§4.7); the fusion thread re-asserts both on every sample, so a config reload cannot leave the skip band and `Ra_scale` disagreeing.

Source: EMA threshold detector with Schmitt hysteresis, standard (*canonical*).

10. Startup gyro-bias estimation — `fusion_thread()` (`imu.c:856`)

Mean of the gyro over a still window ($N = \text{gyro_bias_sec} \cdot \text{ODR}$ samples):

$$\hat{b}_k = \frac{1}{N} \sum_{i=1}^N \omega_{i,k}.$$

A per-axis motion check computes the sample standard deviation $\sigma_k = \sqrt{\omega^2 - \bar{\omega}^2}$; if $\max_k \sigma_k > 0.00873 \text{ rad} \cdot \text{s}^{-1}$ ($0.5^\circ \cdot \text{s}^{-1}$) the window is **doubled once** (a longer average spans more wave cycles). The result seeds `mekf_init` (§4.2); the MEKF refines it online regardless.

Source: sample mean/variance, elementary (*canonical*).

11. Timestamp anchoring and per-sample dt — `imu_math.c:58–194`

A single anchor (`chip_ticks`, `wall_ns`, `tai_ns`, `gen`) maps the sensor's free-running counter to system time. Per sample (`chip_to_wall`, `imu_math.c:178`), with `tick_ns` the counter period:

$$\text{offset} = (\text{chip_ts} - \text{chip_ticks}) \cdot \text{tick_ns}, \quad t_{\text{wall}} = \text{wall_ns} + \text{offset},$$

t_{tai} likewise. Subtraction is unsigned 32-bit (wrap-safe within one anchor interval). The anchor is refreshed on first read, every 60 s with a hardware counter, or every burst without one.

Per-sample interval for prediction (`imu.c`): the wall-time delta between consecutive samples of the same anchor generation, **clamped to** $[0.5, 2] \times \text{nominal}$; outside that (FIFO gap, anchor reset) the nominal 1/ODR is used. This keeps oscillator tolerance (few % on the chip clock) from scaling integrated rotation while rejecting gap artefacts.

As-implemented. Linear (constant-rate) interpolation between anchors; no clock-drift model beyond the 60 s re-anchor. `tick_ns=0` for chips without a hardware timestamp degenerates the offset to 0 (the anchor wall-time is then the per-sample time).

Source: imud-specific timestamp reconstruction.

12. Offline calibration fits (`imud-cal`, `cal_math.c`)

12.1 Linear solver — `gauss4()` (`cal_math.c:25`)

Gaussian elimination with **partial pivoting** on the augmented 4×5 system, then back-substitution. Returns -1 if any pivot $< 10^{-12}$. Double precision.

Source: Golub & Van Loan 2013 §3.4 (*canonical*).

12.2 Hard-iron sphere fit — `sphere_fit()` (`cal_math.c:153`)

Algebraic (linearized) sphere fit. The sphere $(x - c_x)^2 + (y - c_y)^2 + (z - c_z)^2 = r^2$ is linearized as

$$x^2 + y^2 + z^2 = 2c_x x + 2c_y y + 2c_z z + (r^2 - \|c\|^2),$$

and the normal equations $A^\top A p = A^\top b$ are accumulated incrementally (`sphere_add`, `cal_math.c:59`) over sums $\Sigma x, \Sigma x^2, \Sigma xy, \Sigma xr^2, \dots$. Solved by `gauss4`:

$$A^\top A = \begin{bmatrix} 4S_{xx} & 4S_{xy} & 4S_{xz} & 2S_x \\ 4S_{xy} & 4S_{yy} & 4S_{yz} & 2S_y \\ 4S_{xz} & 4S_{yz} & 4S_{zz} & 2S_z \\ 2S_x & 2S_y & 2S_z & n \end{bmatrix}, \quad A^\top b = \begin{bmatrix} 2S_{xr} \\ 2S_{yr} \\ 2S_{zr} \\ S_{xx} + S_{yy} + S_{zz} \end{bmatrix},$$

center $c = p[0:3]$, $r = \sqrt{p_3 + \|c\|^2}$. Center = hard-iron offset; r = reference field magnitude for §12.3.

As-implemented. This is the *algebraic* fit (minimizes an algebraic residual, not orthogonal geometric distance); fast, closed-form, and adequate for well-sampled swings, but statistically biased for sparse/noisy data versus a geometric fit.

Source: algebraic sphere/magnetometer fit, Renaudin et al. 2010; Coope 1993 (*canonical*).

12.3 Soft-iron 2-D ellipse fit — `ellipse_fit()` (`cal_math.c:106`)

Conic least-squares on the horizontal magnetometer locus $Ax^2 + Bxy + Cy^2 = 1$ (regressors x^2, xy, y^2 ; normal equations from $S_{40}, S_{31}, S_{22}, S_{13}, S_{04}, S_{20}, S_{11}, S_{02}$ via `ellipse_add`, solved by `gauss4` with an identity pad row). Requires the conic $M = \begin{pmatrix} A & B/2 \\ B/2 & C \end{pmatrix}$ positive-definite ($A > 0, \det > 0$).

Soft-iron matrix $S = r \cdot M^{1/2}$ via the 2×2 symmetric eigendecomposition: eigenvalues $\lambda_{1,2} = \frac{A+C}{2} \pm \sqrt{\frac{1}{4}(A-C)^2 + \frac{1}{4}B^2}$, unit eigenvector v for λ_1 , and

$$S = s_1 v v^\top + s_2 v_\perp v_\perp^\top, \quad s_{1,2} = r \sqrt{\lambda_{1,2}}, \quad v_\perp = (-v_y, v_x).$$

As-implemented / audit note. The soft-iron correction is **2-D** (X–Y plane only). This is a deliberate specialization for the heading-only fusion default (§4.8): the swing is a horizontal circle, so the vertical/dip

channel is not calibrated. `ellipse_fit` includes a diagonal fallback when the conic is degenerate. The full `mag_soft_iron[3][3]` applied live (§3.3) therefore has its Z couplings from configuration/identity, not from this fit.

Source: algebraic conic/ellipse fit, Fitzgibbon, Pilu & Fisher 1999; magnetometer soft-iron, Renaudin et al. 2010 (*canonical*).

12.4 Heading-circle coverage — `cal_cov_mark()` (`cal_math.c:308`)

Guided-swing coverage: the heading circle is split into `nsec` sectors; the sample (x, y) relative to center (c_x, c_y) is binned by

$$s = \left\lfloor \frac{\text{atan2}(y - c_y, x - c_x) \bmod 2\pi}{2\pi/\text{nsec}} \right\rfloor.$$

`cal_cov_count` sums the marked sectors. Pure bookkeeping (no fit).

12.5 Allan variance — `allan_deviation()` (`cal_math.c:327`)

Overlapping Allan deviation. With the cumulative integral $\theta[k] = \sum_{j < k} x_j dt$ and octave cluster lengths $m = 1, 2, 4, \dots$ ($\tau = m dt$):

$$\sigma^2(\tau) = \frac{1}{2\tau^2 (N - 2m + 1)} \sum_{k=0}^{N-2m} (\theta[k+2m] - 2\theta[k+m] + \theta[k])^2.$$

Characterization (`allan_characterize`, `cal_math.c:360`): white-noise density from the shortest cluster,

$$N = \sigma(\tau_0) \sqrt{\tau_0},$$

bias instability from the curve minimum,

$$B = 0.664 \cdot \min_{\tau} \sigma(\tau).$$

As-implemented. Octave (not fully-overlapping all- τ) spacing; N is read at τ_0 (assumes white noise dominates there); B uses the IEEE-952 factor 0.664 at the deviation minimum (an upper bound if the flat region is not reached). These characterize the sensor; they do **not** feed the filter tuning (§4.2 note).

Source: Allan 1966; overlapping estimator Riley 2008 (NIST SP 1065); N/B extraction and 0.664, IEEE Std 952 (*code comment: "IEEE 952"*).

12.6 Gyro-bias / temperature fit — `gyro_temp_fit()` (`cal_math.c:384`)

Ordinary least-squares line of gyro bias vs. temperature about a reference T_{ref} : with $t_i = T_i - T_{ref}$,

$$\text{coeff} = \frac{n \sum t_i y_i - \sum t_i \sum y_i}{n \sum t_i^2 - (\sum t_i)^2}, \quad \text{bias_ref} = \frac{\sum y_i - \text{coeff} \sum t_i}{n}.$$

Rejects fits with temperature span < 1 °C. `coeff` $\rightarrow$ `gyro_temp_coeff`, applied live in §3.2.

Source: ordinary least squares, elementary (*canonical*).

13. World Magnetic Model — `wmm_field_ned()` (`wmm.c:119`)

Degree/order 12 spherical-harmonic geomagnetic field. This section states the implemented computation; the defining theory is the WMM report cited below.

Secular variation to the decimal year t (`wmm_decimal_year`, `wmm.c:73`), $\Delta t = t - t_{epoch}$:

$$g_n^m(t) = g_n^m + \dot{g}_n^m \Delta t, \quad h_n^m(t) = h_n^m + \dot{h}_n^m \Delta t.$$

Geodetic (WGS-84) → geocentric spherical. With prime-vertical radius $N = a/\sqrt{1 - e^2 \sin^2 \varphi}$ ($a = 6378137$ m, e^2 from $f = 1/298.257223563$):

$$p_x = (N + \text{alt}) \cos \varphi, \quad p_z = (N(1 - e^2) + \text{alt}) \sin \varphi,$$

$$r = \sqrt{p_x^2 + p_z^2}, \quad \theta = \arccos(p_z/r) \text{ (co-latitude)}, \quad \lambda = \text{lon}.$$

Schmidt quasi-normalized associated Legendre functions P_n^m and derivatives $dP_n^m/d\theta$, as coded (`wmm.c:178`): seeds $P_0^0 = 1$, $P_1^0 = \cos \theta$, $P_1^1 = \sin \theta$; for $n \geq 2$,

$$P_n^n = \sqrt{\frac{2n-1}{2n}} \sin \theta P_{n-1}^{n-1}, \quad P_n^m = \frac{2n-1}{\sqrt{n^2 - m^2}} \cos \theta P_{n-1}^m - \sqrt{\frac{(n-m-1)(n+m-1)}{(n-m)(n+m)}} P_{n-2}^m,$$

the second covering zonal ($m = 0$), sub-diagonal ($m = n - 1$, second term 0), and tesseral terms; derivatives follow by differentiating the same recursion.

Field synthesis (geocentric NED), $a = 6371.2$ km:

$$X_{gc} = \sum_{n=1}^{12} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^n (g_n^m \cos m\lambda + h_n^m \sin m\lambda) \frac{dP_n^m}{d\theta},$$

$$Y_{gc} = \frac{1}{\sin \theta} \sum_{n=1}^{12} \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^n m (g_n^m \sin m\lambda - h_n^m \cos m\lambda) P_n^m,$$

$$Z_{gc} = - \sum_{n=1}^{12} (n+1) \left(\frac{a}{r}\right)^{n+2} \sum_{m=0}^n (g_n^m \cos m\lambda + h_n^m \sin m\lambda) P_n^m.$$

Geocentric → geodetic NED rotation by $\delta = \varphi - \varphi_{gc}$ ($\varphi_{gc} = \pi/2 - \theta$):

$$X_{NED} = X_{gc} \cos \delta + Z_{gc} \sin \delta, \quad Y_{NED} = Y_{gc}, \quad Z_{NED} = Z_{gc} \cos \delta - X_{gc} \sin \delta.$$

Declination (`wmm_declination`, `wmm.c:251`): $D = \text{atan2}(Y_{NED}, X_{NED})$. The horizontal magnitude $H = \sqrt{X^2 + Y^2}$ and vertical Z supply the §4.9 magnetic invariants.

As-implemented. Secular variation is linear extrapolation from the epoch coefficients (valid over the model's 5-year validity window). `wmm_load` requires exactly the 90 degree-12 coefficient rows and rejects truncated files. Powers $(a/r)^{n+2}$ are formed by iterative multiplication, not `pow`. Poles are guarded by clamping $\sin \theta \geq 10^{-10}$.

Source: WMM2025 Technical Report, Chulliat et al. 2024 (NCEI/BGS) — the defining document; ALF recursion Langel 1987 / WMM Technical Note 28 (*code comment*); WGS-84 NIMA TR8350.2 (*canonical*).

14. Symbol ↔ code map

Symbol	Meaning	Code
q	attitude quaternion, body→NED, scalar-first	<code>mekf_t.q[4]</code>
b	gyro bias	<code>mekf_t.bias[3]</code>
a_w	wave acceleration, body, g units	<code>mekf_t.wave_acc[3]</code>
P	9×9 error covariance $[\delta\theta \mid \delta b \mid \delta a_w]$	<code>mekf_t.P[MEKF_N][MEKF_N]</code>
$R(q)$	body→NED rotation matrix	<code>q_to_R()</code>
Q_g, Q_b	process-noise variances / step	<code>mekf_t.Qg, .Qb</code>
R_a, R_m	accel / mag meas. noise var	<code>mekf_t.Ra, .Rm</code>
σ, τ	Gauss–Markov wave σ ($\text{m} \cdot \text{s}^{-2}$), correlation time (s)	<code>mekf_wave_accel, mekf_wave_accel_tau_s</code>
σ_g^2, φ, Q_w	wave variance (g^2), GM transition, GM noise	<code>mekf_t.wave_sig2, mekf_predict</code>
$P_{\hat{z}}$	tangent projector $I - \hat{z}\hat{z}^\top$	<code>wave_jacobian</code>
H	measurement Jacobian	<code>eskf_update, eskf_update_yaw</code>
γ	χ^2 innovation gate	<code>ACCEL_CHI2_GATE=11.34, YAW_CHI2_GATE=6.63</code>
m_{ref}	Earth-field reference, NED, Gauss	<code>mekf_t.m_ref[3]</code>
q_{quiet}	quiescence EMA \$(	<code>a</code>
g_{body}	accel gravity direction, body	<code>mekf_t.g_body[3]</code>
H_s, T_z	sig. wave height, zero-cross period	<code>seastate_wave_height/_period</code>
a_D	NED-down linear accel	<code>heave_update</code>
N, B	Allan noise density, bias instability	<code>allan_characterize</code>
g_n^m, h_n^m	Gauss coefficients	<code>wmm_t.g/.h, .g_sv/.h_sv</code>
D	magnetic declination	<code>wmm_declination</code>

15. References

Citations are to the canonical source for each method; `imud` implements standard techniques rather than novel mathematics. Items the maintainer may wish to verify against a specific edition/page are flagged `[verify]`.

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*This document reflects the code at the release it ships with. When a numerical routine changes, update the corresponding section and its **As-implemented** note in the same commit.*